

QECTOR Workbench - Complete API Reference

****Workbench 3.5.1 - Backend qector_decoder_v3 0.6.9 (min 0.6.2) - 47 MCP tools - 13 decoders - 9 code families****

****PyPI package: [qector-decoder-v3](<https://pypi.org/project/qector-decoder-v3/>) 0.6.9****

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This manual is generated from the live application source so every tool name, decoder kind, code family, and function signature matches the running build exactly.

Code families

Family	Parameter	Type	Graphlike	Decoders	Notes
repetition	distance	int	yes	all (13)	1D chain parity-check code.
ring	distance	int	yes	all (13)	Periodic 1D chain.
rotated_surface	distance	int	yes	all (13)	Standard rotated surface code.
unrotated_surface	distance	int	yes	12 (lookup_table refused >20 checks)	Square lattice surface code.
toric	distance	int	yes	12 (lookup_table refused >20 checks)	Toric code with periodic boundaries.
heavy_hex	distance	int	yes	all (13)	IBM heavy-hex lattice.
hypergraph_product	distance	int	yes	all (13)	CSS from repetition seed; graphlike.
bicycle	circulant size	int	no	all (13)	qLDPC bicycle code; graphlike enough for all decoders.
bivariate_bicycle	preset index	int	no	9 (excludes union_find, fast_union_find, lookup_table, belief_propagation)	

Decoder kinds

Kind	Description	Options	Compatibility
union_find	Fast approximate matching via union-find.	bp_method, osd_order ignored	graphlike only
fast_union_find	Faster union-find variant; approximate, higher LER.	-	graphlike only
blossom	Weight-optimal exact MWPM; matches PyMatching LER.	-	all
sparse_blossom	Region-growing near-optimal matching; not exact.	-	graphlike only
bp_osd	Belief propagation + ordered statistics for LDPC/qLDPC.	bp_method, osd_order, error_rate	all
auto	Self-selecting AutoDecoder.	-	graphlike only
hybrid	Combines multiple strategies; chooses per problem.	-	graphlike only
lookup_table	Exhaustive syndrome-to-correction table; refused above 20 checks.	-	small codes only
predecoded	Fast pre-decoding pass before matching.	-	graphlike only
auto_router	Policy decoder: matching for graphlike, bp_osd for qLDPC. Universally compatible.	-	all
hybrid_cascade	Union-Find pre-filter + Blossom/BP-OSD escalation; exposes cascade stats.	escalation, err	
gnn_belief_matching	GNN-guided weighted matching with faithfulness fallback.	gnn_hidden_size, gnn_n_laye	
belief_matching	BP posteriors reweight exact Blossom matching; faithfulness fallback.	bp_method, osd_ord	

Decoder options

Key	Type	Applies to	Description
bp_method	string	bp_osd, belief_matching	"exact" or "min_sum".
osd_order	int	bp_osd, belief_matching	0, 1, or 2. Higher is slower/more accurate.
error_rate	float	all	Physical error probability used to weight edges or set BP priors.
escalation	string	hybrid_cascade	"blossom" or "bp_osd".
max_accept_weight	int	hybrid_cascade	Maximum syndrome weight accepted by the pre-filter.
gnn_hidden_size	int	gnn_belief_matching	Hidden dimension of the GNN.
gnn_n_layers	int	gnn_belief_matching	Number of GNN message-passing layers.

Unknown keys are ignored with a warning; missing keys use backend defaults.

Measured data

All figures below were measured on this machine (seeded, n=50, p=0.05, rotated_surface d=5). They are workload- and hardware-dependent.

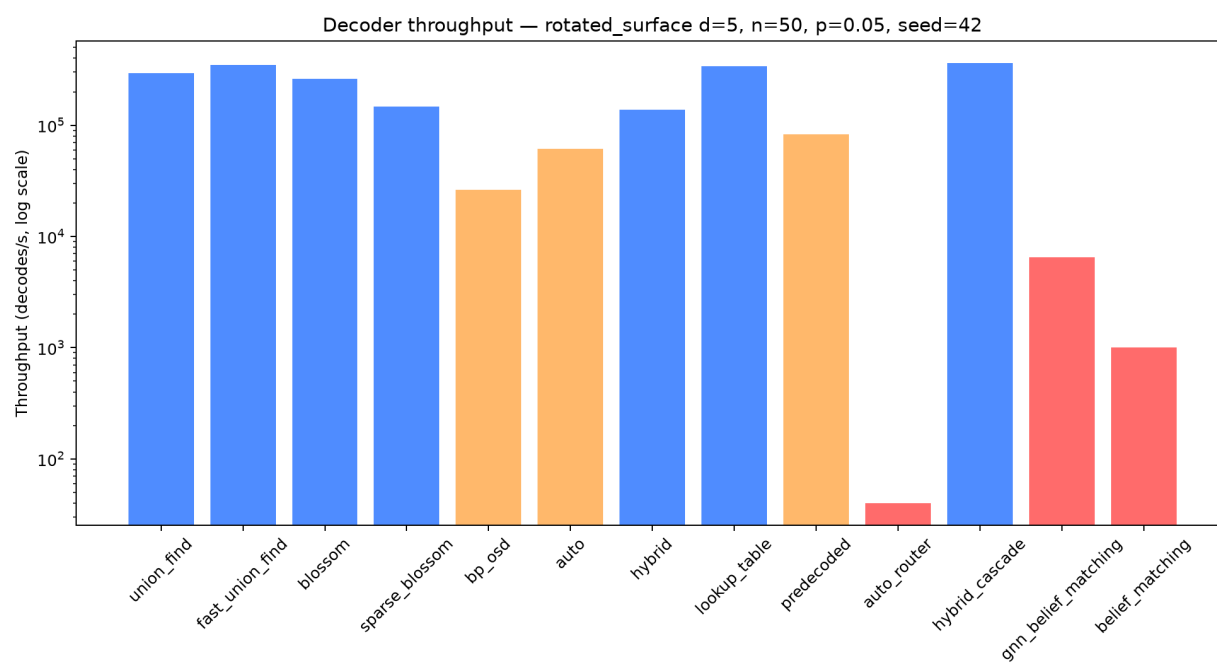
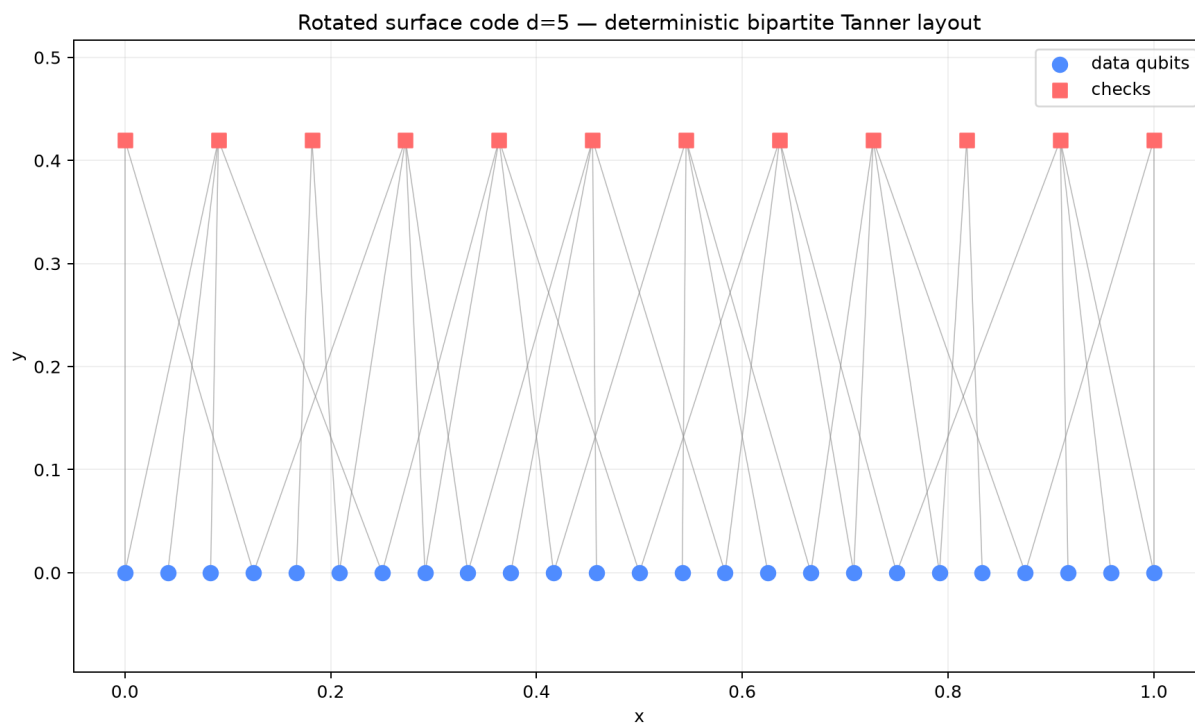
Code family properties

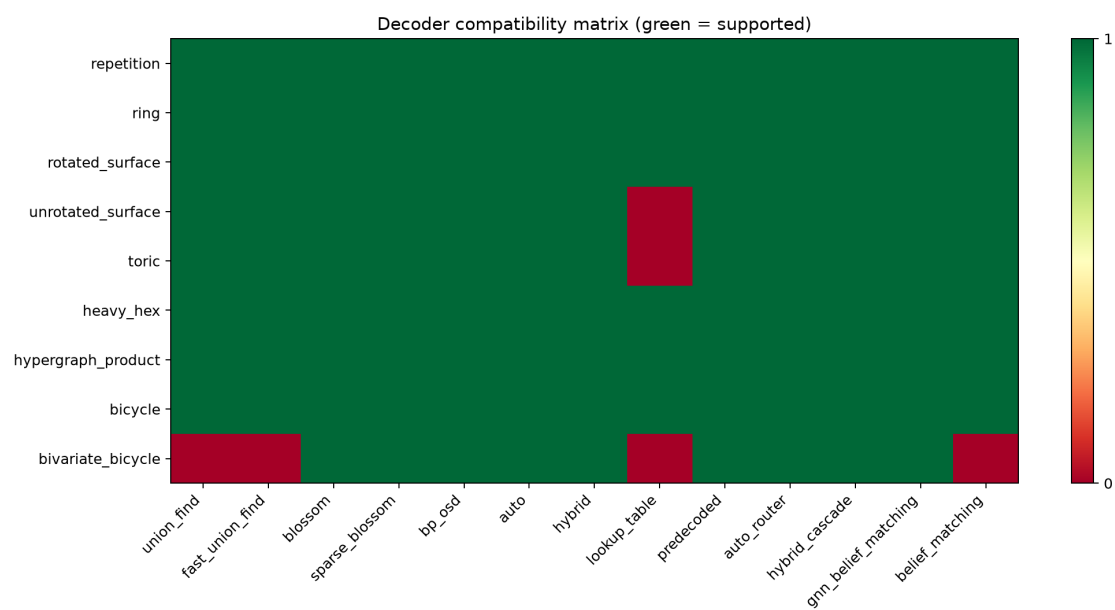
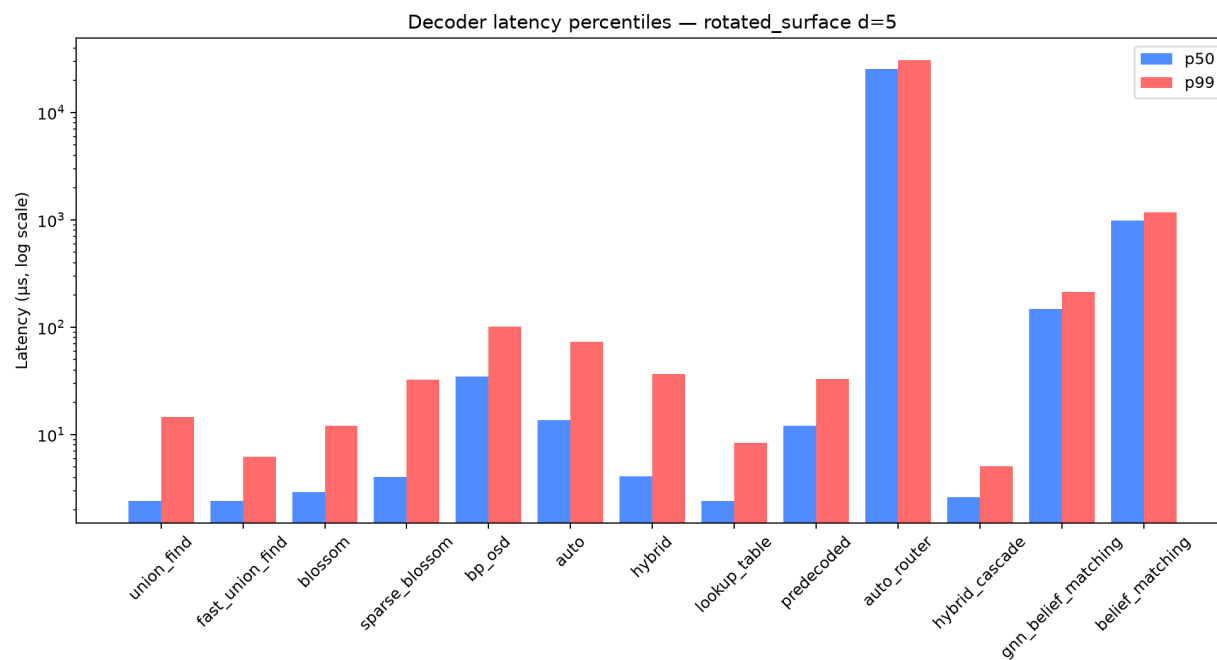
Family	Distance	n_qubits	n_checks	max_degree	compatible decoders
repetition	5	5	4	2	13
ring	5	5	5	2	13
rotated_surface	5	25	12	2	13
unrotated_surface	5	40	25	2	12
toric	5	50	25	2	12
heavy_hex	5	25	20	2	13
hypergraph_product	5	41	20	2	13
bicycle	5	10	5	2	13
bivariate_bicycle	3	72	36	3	9

Decoder benchmark results

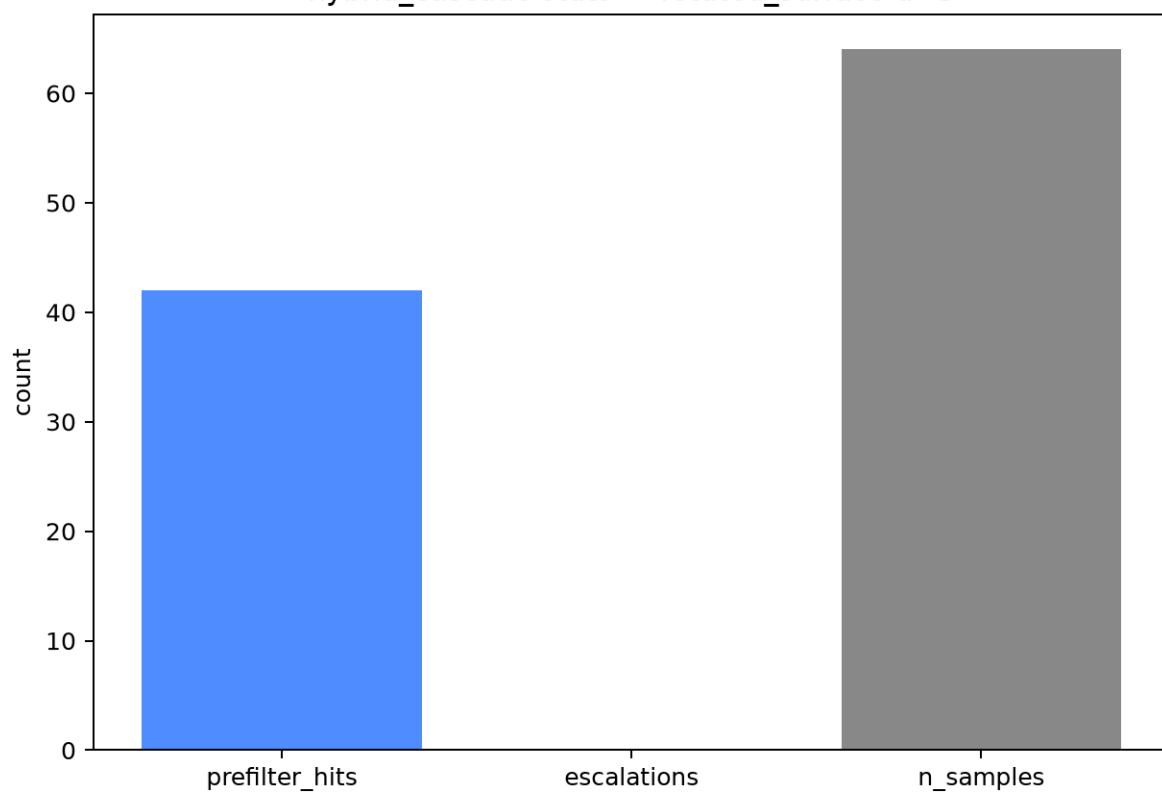
Decoder	Throughput (decodes/s)	p50 latency (μs)	p99 latency (μs)	logical error rate
union_find	295508	2.40	14.54	0.1
fast_union_find	349895	2.40	6.25	0.1
blossom	261917	2.90	12.02	0.08
sparse_blossom	146757	4.05	32.63	0.08
bp_osd	26162	34.75	101.55	0.1
auto	61125	13.60	73.22	0.1
hybrid	138812	4.10	36.70	0.08
lookup_table	337610	2.40	8.43	0.1
predecoded	82850	12.00	32.98	0.08
auto_router	40	25457.50	30758.21	0.08
hybrid_cascade	362845	2.60	5.05	0.1
gnn_belief_matching	6520	147.15	213.77	0.08
belief_matching	1001	988.05	1173.28	0.02

Figures





hybrid_cascade stats — rotated_surface d=5



backend.py API

``backend.Any(*args, **kwargs)``

Special type indicating an unconstrained type.

``backend.Optional(*args, **kwargs)``

Optional[X] is equivalent to Union[X, None].

``backend.QectorError(...)``

Raised for invalid operations in the QECTOR backend.

``backend.build_code(family_key: 'str', param: 'int')``

Build a code from a family and parameter (distance).

``backend.code_summary(code) -> 'dict[str, Any]``

Return a summary dict for a code object.

``backend.compat_report() -> 'dict[str, Any]``

Ecosystem-integration availability report.

``backend.compatible_decoder_kinds(code) -> 'list[str]``

Return the decoder kinds that can actually construct on code.

``backend.compute_spring_layout(n_qubits: 'int', n_checks: 'int', check_matrix, iterations: 'int' = 100) -> 'tuple[list, list]``

Backwards-compatible alias for :func:compute_tanner_layout.

``backend.compute_tanner_layout(n_qubits: 'int', n_checks: 'int', check_matrix) -> 'tuple[list, list]``

Deterministic bipartite Tanner-graph layout.

``backend.deque(...)``

deque([iterable[, maxlen]]) --> deque object

`backend.get\_code\_family\_info(family\_key: 'str') -> 'dict[str, str]`

Return metadata about a code family.

`backend.get\_compatible\_decoders(code) -> 'list[dict[str, str]]`

Return decoder info for the decoders that can construct on this code.

`backend.get\_decoder\_info(kind: 'str') -> 'dict[str, str]`

Return human-readable info about a decoder kind.

`backend.get\_tanner\_graph\_layout(code, family: 'str', distance: 'int') -> 'tuple[list, list]`

Return qubit and check coordinates for a clean bipartite Tanner graph.

`backend.list\_available\_codes() -> 'dict[str, Any]`

Code families wired into the workbench plus the backend's native codes.list_codes() catalogue (v0.6.6). Pure introspection.

`backend.logical\_failure(logicals: 'np.ndarray', error, correction) -> 'bool`

Public: True iff residual (error+correction)%2 flips a logical.

`backend.logicals\_matrix(code) -> 'Optional[np.ndarray]`

Public accessor for the code's logical-operator matrix (or None).

`backend.make\_decoder(code, decoder\_kind: 'str', decoder\_options: 'Optional[dict]' = None) -> 'Any`

Public: construct a decoder of decoder_kind for code.

`backend.native\_recommend(family\_key: 'Optional[str]' = None, distance: 'Optional[int]' = None, n\_qubits: 'Optional[int]' = None, priority: 'str' = 'balanced', batch\_size: 'int' = 1) -> 'dict[str, Any]`

Backend-native decoder recommendation (v0.6.6 recommend).

`backend.run\_batch\_decode(code, backend: 'str' = 'cpu', n\_samples: 'int' = 100, error\_rate: 'float' = 0.05, seed: 'int' = 1) -> 'dict[str, Any]`

Run a batch decode on the given code.

```
`backend.run_benchmark(code, n_samples: 'int' = 1000, seed: 'int' = 42, decoder_kind: 'str' = 'union_find', error_rate: 'float' = 0.05) -> 'dict[str, Any]`
```

Run a decode benchmark on the given code.

```
`backend.run_diagnostic_decode(code, error_rate: 'float' = 0.05, decoder_kind: 'str' = 'blossom', seed: 'int' = 42) -> 'dict[str, Any]`
```

Rich single decode via v0.6.6 decode_with_diagnostics.

```
`backend.run_hybrid_cascade_stats(code, n_samples: 'int' = 64, error_rate: 'float' = 0.05, seed: 'int' = 1, escalation: 'Optional[str]' = None) -> 'dict[str, Any]`
```

Batch-decode with HybridCascadeDecoder and expose its cascade statistics.

```
`backend.run_native_streaming(code, n_rounds: 'int' = 8, error_rate: 'float' = 0.03, seed: 'int' = 1, window_size: 'int' = 4) -> 'dict[str, Any]`
```

Native sliding-window streaming decode (v0.6.6 sliding_window_decode).

```
`backend.run_neural_predecoder_training(code, n_samples: 'int' = 200, n_epochs: 'int' = 5, error_rate: 'float' = 0.05, seed: 'int' = 1) -> 'dict[str, Any]`
```

Train the NeuralPredecoder on sampled (syndrome, error) pairs (lab tool).

```
`backend.run_parallel_batch_decode(code, n_samples: 'int' = 64, error_rate: 'float' = 0.05, seed: 'int' = 1, decoder_type: 'str' = 'union_find', n_workers: 'Optional[int]' = None) -> 'dict[str, Any]`
```

Multi-process parallel batch decode via v0.6.6 DecoderPool.

```
`backend.run_single_decode(code, error_rate: 'float', decoder_kind: 'str', seed: 'int', decoder_options: 'Optional[dict]' = None) -> 'dict[str, Any]`
```

Run a single decode on a code with a random error.

```
`backend.run_streaming_session(code, window_size: 'int' = 5, n_rounds: 'int' = 10, error_rate: 'float' = 0.03, seed: 'int' = 1, decoder_kind: 'str' = 'union_find') -> 'dict[str, Any]`
```

Run a sliding-window streaming decode session.

```
`backend.sample_error_and_syndrome(code, error_rate: 'float', seed: 'int')`
```

Public: sample one seeded error and its syndrome for code.

```
`backend.validate_parameter(family_key: 'str', param: 'int') -> 'tuple[bool, str]`
```

Validate a code family parameter (distance).

```
`backend.verify_correction(code, syndrome, correction) -> 'bool`
```

Public: True iff correction reproduces the observed syndrome.

autodebug.py API

```
`autodebug.Optional(*args, **kwds)`
```

Optional[X] is equivalent to Union[X, None].

```
`autodebug.asdict(obj, *, dict_factory=)`
```

Return the fields of a dataclass instance as a new dictionary mapping field names to field values.

```
`autodebug.dataclass(cls=None, /, *, init=True, repr=True, eq=True, order=False, unsafe_hash=False, frozen=False, match_args=True, kw_only=False, slots=False, weakref_slot=False)`
```

Add dunder methods based on the fields defined in the class.

```
`autodebug.field(*, default=, default_factory=, init=True, repr=True, hash=None, compare=True, metadata=None, kw_only=)`
```

Return an object to identify dataclass fields.

```
`autodebug.probe_decoders(family: 'str', distance: 'int', error_rate: 'float' = 0.05, seed: 'int' = 42, verify: 'bool' = True) -> 'dict`
```

Try every wired decoder against one seeded syndrome and report results.

```
`autodebug.resilient_batch_decode(family: 'str', distance: 'int', backend: 'str' = 'cuda', n_samples: 'int' = 100, error_rate: 'float' = 0.05, seed: 'int' = 1, fallback_chain: 'Optional[list]' = None) -> 'dict`
```

Batch-decode, falling back cuda → opencl → cpu on unavailability.

```
`autodebug.resilient_single_decode(family: 'str', distance: 'int', error_rate: 'float' = 0.05,
decoder: 'str' = 'union_find', seed: 'int' = 42, fallback_chain: 'Optional[list]' = None, verify:
'bool' = True) -> 'ResilientDecodeResult`
```

Decode one seeded syndrome, falling back through decoders on failure.

```
`autodebug.run_self_diagnostics(probe_family: 'str' = 'repetition', probe_distance: 'int' =
3) -> 'DiagnosticsReport`
```

Run a full environment/decoder/hardware self-test.

```
`autodebug.self_diagnostics(probe_family: 'str' = 'repetition', probe_distance: 'int' = 3) ->
'DiagnosticsReport`
```

Run a full environment/decoder/hardware self-test.

hardware_routing.py API

```
`hardware_routing.Optional(*args, **kwds)`
```

Optional[X] is equivalent to Union[X, None].

```
`hardware_routing.dataclass(cls=None, /, *, init=True, repr=True, eq=True, order=False,
unsafe_hash=False, frozen=False, match_args=True, kw_only=False, slots=False,
weakref_slot=False)`
```

Add dunder methods based on the fields defined in the class.

```
`hardware_routing.detect_hardware() -> 'HardwareProfile`
```

Detect real GPU/CUDA/OpenCL availability via the installed package.

```
`hardware_routing.recommend(code_family: 'Optional[str]', distance: 'Optional[int]',
n_qubits: 'Optional[int]', priority: 'str') -> 'Recommendation`
```

Heuristic decoder recommendation (deterministic, no model call).

version_service.py API

```
`version_service.Callable(*args, **kwargs)`
```

Deprecated alias to `collections.abc.Callable`.

``version_service.Optional(*args, **kwargs)``

`Optional[X]` is equivalent to `Union[X, None]`.

``version_service.effective_app_version(prefer_latest: 'bool' = False) -> 'Optional[str]``

The version the app presents as *its own* — it tracks the decoder backend.

``version_service.format_version_banner(report: 'Optional[dict]' = None) -> 'str``

One-line human banner where the app version *is* the decoder version.

``version_service.get_app_version_info(refresh: 'bool' = False) -> 'dict[str, Any]``

Local baseline vs. latest-on-PyPI for the workbench application.

``version_service.get_backend_version_info(refresh: 'bool' = False) -> 'dict[str, Any]``

Installed vs. latest-on-PyPI for the decoder backend.

``version_service.get_version_report(refresh: 'bool' = False) -> 'dict[str, Any]``

Combined app + backend version report resolved from PyPI (cached).

``version_service.installed_backend_version() -> 'Optional[str]``

The version of the compiled decoder backend actually imported, if any.

``version_service.is_newer(latest: 'Optional[str]', current: 'Optional[str]') -> 'bool``

True iff latest is a strictly newer version than current.

``version_service.local_app_version() -> 'str``

The workbench's baked-in baseline version (offline-safe).

``version_service.parse_version(v: 'Optional[str]') -> 'tuple``

Parse a version string to a tuple of ints for ordering.

``version_service.resolve_versions_async(callback: 'Optional[Callable[[dict], None]]' = None, refresh: 'bool' = False) -> 'threading.Thread``

Resolve both versions on a daemon thread, then invoke `callback(report)`.

decoder_provisioner.py API

``decoder_provisioner.Callable(*args, **kwargs)``

Deprecated alias to `collections.abc.Callable`.

``decoder_provisioner.Optional(*args, **kwds)``

`Optional[X]` is equivalent to `Union[X, None]`.

``decoder_provisioner.abi_tag() -> 'str'``

A short tag unique to this interpreter's binary ABI.

``decoder_provisioner.activate_site() -> 'Optional[Path]``

Place the active managed site first on `sys.path` (idempotently).

``decoder_provisioner.active_site() -> 'Optional[Path]``

Read the active managed-site pointer; malformed pointers are ignored.

``decoder_provisioner.bootstrap(on_log: 'Optional[Callable[[str], None]]' = None) -> 'dict'``

Blocking pre-import gate used by `main.py` before backend imports.

``decoder_provisioner.ensure(prefer_latest: 'bool' = True, timeout: 'int' = 300, on_log: 'Optional[Callable[[str], None]]' = None, target_version: 'Optional[str]' = None) -> 'dict'``

Ensure an importable decoder, optionally installing a newer PyPI wheel.

``decoder_provisioner.ensure_async(prefer_latest: 'bool' = True, callback: 'Optional[Callable[[dict], None]]' = None, on_log: 'Optional[Callable[[str], None]]' = None, target_version: 'Optional[str]' = None) -> 'threading.Thread'``

Run `:func:ensure` on a daemon thread; upgrade takes effect next launch.

``decoder_provisioner.ensure_dependencies(on_log: 'Optional[Callable[[str], None]]' = None) -> 'dict'``

Check and automatically install any missing core dependencies via `pip`.

`decoder\_provisioner.import\_ok() -> 'bool'`

True iff the decoder *actually imports* in this interpreter — i.e. its compiled extension loads. A metadata-only presence check is not enough: a wheel built for another Python ABI leaves valid dist-info but an unloadable .pyd/.so. This is the authoritative "is a usable decoder present?" test used by the boot gate.

`decoder\_provisioner.is\_frozen() -> 'bool'`

No docstring.

`decoder\_provisioner.managed\_root() -> 'Path'`

Return the app-owned, user-writable, ABI-scoped decoder storage dir.

`decoder\_provisioner.resolve\_pip\_argv() -> 'tuple[Optional[list[str]], str]'`

Find a pip interpreter that can install extension modules for this app.

`decoder\_provisioner.scan\_version() -> 'Optional[str]'`

Return the active managed version, otherwise a system-installed version.

`decoder\_provisioner.self\_check() -> 'dict'`

No docstring.

auto_updater.py API

`auto\_updater.Optional(\*args, \*\*kwds)`

Optional[X] is equivalent to Union[X, None].

`auto\_updater.auto\_upgrade\_on\_boot(callback=None) -> 'dict'`

Keep the app on the latest live PyPI decoder backend, automatically.

`auto\_updater.boot\_update\_summary(refresh: 'bool' = False) -> 'dict'`

Combined app + backend update summary for boot / on-demand checks.

`auto\_updater.check\_app\_update() -> 'Optional[str]'`

Latest workbench app version on PyPI if newer than the local build, else None.

`auto\_updater.check\_for\_update() -> 'Optional[str]`

Check PyPI for a newer version. Returns version string if update available, else None.

`auto\_updater.perform\_backend\_update(callback=None, target\_version: 'Optional[str]' = None) -> 'dict`

Push (install) the latest decoder backend from PyPI in the background.

`auto\_updater.try\_upgrade(target\_version: 'str', callback=None) -> 'None`

Upgrade in a background thread, optionally calling callback(success, msg).

`auto\_updater.upgrade\_instructions(latest: 'str') -> 'str`

Return upgrade guidance appropriate to how the app was installed.

doc_generator.py API

`doc\_generator.Optional(\*args, \*\*kwds)`

Optional[X] is equivalent to Union[X, None].

`doc\_generator.latex\_escape(value: 'Any') -> 'str`

Escape LaTeX special characters (\ & % \$ # _ { } ~ ^) in value.

MCP tool reference

47 tools via stdio JSON-RPC 2.0.

``analyze_code_family``

Analyze a code family with an example code instance

****Parameters****

- family_name (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -

``batch_decode``

Batch-decode sampled syndromes on cpu/cuda/opengl via backend.run_batch_decode

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- backend (string, default 'cpu') - One of: cpu, cuda, opengl (no silent fallback)
- n_samples (integer, default 100) -
- error_rate (number, default 0.05) -
- seed (integer, default 1) -

``batch_decode_gpu``

Batch-decode on an explicit compute backend (cpu/cuda/opengl) with honest availability reporting — unavailable GPU backends return status='unavailable' with a reason, never fake results

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 3) -
- backend (string, default 'cuda') - One of: cpu, cuda, opengl
- n_samples (integer, default 32) -
- error_rate (number, default 0.05) -
- seed (integer, default 1) -

``belief_match_decode``

Convenience seeded decode pinned to the 0.6.9 belief_matching kind (BP posteriors + exact MWPM with faithfulness fallback)

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -

- error_rate (number, default 0.05) -
- seed (integer, default 42) -

`benchmark\_decoder`

Benchmark a decoder on a code family via backend.run_benchmark (latency percentiles, throughput, logical error rate)

****Parameters****

- decoder_name (string, default 'union_find') - One of: union_find, fast_union_find, blossom, sparse_blossom, bp_osd, auto, hybrid, lookup_table, predecoded, auto_router, hybrid_cascade, gnn_belief_matching, belief_matching
- code_family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- error_rate (number, default 0.05) -
- n_samples (integer, default 100) -
- seed (integer, default 42) -

`check\_updates`

Check PyPI for newer app / decoder-backend releases and return install-aware upgrade instructions

****Parameters****

- refresh (boolean, default False) - Bypass the version cache and re-query PyPI

`clear\_results`

Clear all stored benchmark results

****Parameters****

- confirm (boolean, default False) -

`compare\_benchmarks`

Compare stored benchmark results side by side (throughput, p99 latency, logical error rate)

****Parameters****

- benchmarks (array, required) - result_id values returned by the run_benchmark tool

`compat\_report`

Report ecosystem-integration availability (stim/sinter/pymatching/qiskit/ldpc) and research components

No parameters.

`compatible\_decoders`

Live probe: which decoder kinds construct and produce a syndrome-verified correction on this code

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 3) -

`decode\_single`

Run one seeded decode and report correction weight, syndrome validity and logical failure

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- decoder_name (string, default 'union_find') - One of: union_find, fast_union_find, blossom, sparse_blossom, bp_osd, auto, hybrid, lookup_table, predecoded, auto_router, hybrid_cascade, gnn_belief_matching, belief_matching
- error_rate (number, default 0.05) -
- seed (integer, default 42) -

`decode\_syndrome`

Decode an explicit 0/1 syndrome (length n_checks) with a chosen decoder; syndrome_valid is the GF(2) re-check, logical_failure is null (no reference error exists, so it is unknowable)

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- decoder_name (string, default 'union_find') - One of: union_find, fast_union_find, blossom, sparse_blossom, bp_osd, auto, hybrid, lookup_table, predecoded, auto_router, hybrid_cascade, gnn_belief_matching, belief_matching
- syndrome (array, required) - 0/1 syndrome bits, length must equal the code's n_checks
- decoder_options ([object, 'null'], required) - Optional per-decoder construction options: bp_method (exact|min_sum), osd_order (0|1|2), error_rate, escalation (blossom|bposd), max_accept_weight, gnn_hidden_size, gnn_n_layers

`decode\_with\_options`

Seeded decode with validated per-decoder construction options (bp_osd bp_method/osd_order, hybrid_cascade escalation, GNN architecture); reports options_applied honestly

****Parameters****

- family (string, default 'repetition') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 3) -
- decoder_name (string, default 'bp_osd') - One of: union_find, fast_union_find, blossom, sparse_blossom, bp_osd, auto, hybrid, lookup_table, predecoded, auto_router, hybrid_cascade, gnn_belief_matching, belief_matching
- error_rate (number, default 0.05) -
- seed (integer, default 42) -
- decoder_options ([object, 'null'], required) - Optional per-decoder construction options: bp_method (exact|min_sum), osd_order (0|1|2), error_rate, escalation (blossom|bposd), max_accept_weight, gnn_hidden_size, gnn_n_layers

`delete\_resource`

Delete a resource by ID

****Parameters****

- resource_id (string, required) -
- confirm (boolean, default False) -

`diagnostic\_decode`

Rich single decode via the backend's native decode_with_diagnostics (matched weight, backend used, internal fallback, timing, logicals)

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- decoder_name (string, default 'blossom') - One of: union_find, fast_union_find, blossom, sparse_blossom, bp_osd, auto, hybrid, lookup_table, predecoded, auto_router, hybrid_cascade, gnn_belief_matching, belief_matching
- error_rate (number, default 0.05) -
- seed (integer, default 42) -

`export\_benchmark`

Export a stored benchmark result (by result_id) to the export directory

****Parameters****

- benchmark_id (string, required) - result_id returned by the run_benchmark tool
- format (string, default 'json') -

`generate\_documentation`

Generate code documentation files

****Parameters****

- family_key (string, default 'ring') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- param (integer, default 6) -
- formats (array, default ['json']) - Any of: json, markdown, html, latex, pdf

`get\_code\_properties`

Get properties of a code family

****Parameters****

- family_name (string, default 'ring') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -

`get\_config`

Get current server configuration

No parameters.

`get\_decoder\_info`

Get information about a decoder

****Parameters****

- decoder_name (string, default 'bp_osd') - One of: union_find, fast_union_find, blossom, sparse_blossom, bp_osd, auto, hybrid, lookup_table, predecoded, auto_router, hybrid_cascade, gnn_belief_matching, belief_matching

`get\_hardware\_info`

Get hardware/backend availability

No parameters.

`get\_resource`

Get a specific resource by ID

****Parameters****

- resource_id (string, required) -

`get\_resources`

List all resources

No parameters.

`get\_results`

Get stored benchmark results (most recent first-in order)

****Parameters****

- limit (integer, default 10) -

`get\_statistics`

Get server statistics

No parameters.

`get\_system\_info`

Get system information

No parameters.

`gnn\_belief\_match\_decode`

Convenience seeded decode pinned to the 0.6.9 gnn_belief_matching kind with optional GNN architecture overrides (gnn_hidden_size, gnn_n_layers)

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product

- distance (integer, default 5) -

- error_rate (number, default 0.05) -

- seed (integer, default 42) -
- gnn_hidden_size (['integer', 'null'], required) -
- gnn_n_layers (['integer', 'null'], required) -

`hybrid\_cascade\_stats`

Batch-decode through the hybrid_cascade decoder and expose its live cascade statistics (prefilter_hits, escalations, hit rate, throughput, syndrome-match rate, logical error rate)

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 3) -
- n_samples (integer, default 64) -
- error_rate (number, default 0.05) -
- seed (integer, default 1) -
- escalation (['string', 'null'], required) - One of: blossom, bposd (default: backend's blossom)

`list\_clients`

List registered clients

No parameters.

`list\_code\_families`

List available code families

No parameters.

`list\_codes`

List workbench code families plus the backend's native code catalogue

No parameters.

`list\_decoders`

List available decoders

No parameters.

`list\_tools`

List all available MCP tools

No parameters.

`mcp\_status`

Get MCP server status

No parameters.

`native\_\_recommend`

Backend-native decoder recommendation (qector_decoder_v3.recommend) with the mapped workbench decoder_kind

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- n_qubits (['integer', 'null'], required) -
- priority (string, default 'balanced') - One of: balanced, speed, accuracy
- batch_size (integer, default 1) -

`native\_\_streaming`

Native hardware-accelerated sliding-window streaming decode (qector_decoder_v3.sliding_window_decode) with per-round validity + telemetry

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- n_rounds (integer, default 8) -
- error_rate (number, default 0.03) -
- seed (integer, default 1) -
- window_size (integer, default 4) -

`neural\_\_predecoder\_\_train`

Train the NeuralPredecoder research/lab MLP on seeded (syndrome, error) pairs and evaluate on a disjoint held-out stream (exact-match, bit accuracy, syndrome validity, LER)

****Parameters****

- family (string, default 'repetition') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 3) -
- n_samples (integer, default 200) -
- n_epochs (integer, default 5) -
- error_rate (number, default 0.05) -
- seed (integer, default 1) -

`probe\_\_decoders`

Probe which decoders produce a valid (syndrome-verified) correction for a code — a self-test across every wired decoder

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product

- distance (integer, default 5) -
- error_rate (number, default 0.05) -
- seed (integer, default 42) -

`recommend\_decoder`

Recommend a decoder for a code/priority using detected hardware

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- n_qubits ([integer, 'null'], required) -
- priority (string, default 'balanced') - One of: balanced, speed, accuracy

`register\_client`

Register a client

****Parameters****

- client_id (string, required) -
- access_level (string, default 'USER') -

`reset\_config`

Reset configuration to defaults

****Parameters****

- confirm (boolean, default False) -

`resilient\_decode`

Single decode with automatic multi-decoder fallback and a full attempt trace (autodebug.resilient_single_decode)

****Parameters****

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- decoder_name (string, default 'union_find') - One of: union_find, fast_union_find, blossom, sparse_blossom, bp_osd, auto, hybrid, lookup_table, predecoded, auto_router, hybrid_cascade, gnn_belief_matching, belief_matching
- error_rate (number, default 0.05) -
- seed (integer, default 42) -

`run\_benchmark`

Run a benchmark and store the result under a generated result_id

****Parameters****

- code_family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product

- distance (integer, default 5) -
- decoder_name (string, default 'union_find') - One of: union_find, fast_union_find, blossom, sparse_blossom, bp_osd, auto, hybrid, lookup_table, predecoded, auto_router, hybrid_cascade, gnn_belief_matching, belief_matching
- n_samples (integer, default 100) -
- seed (integer, default 42) -
- error_rate (number, default 0.05) -

`self\_diagnostics`

Run a full environment/decoder/hardware self-diagnostics report (autodebug.run_self_diagnostics)

No parameters.

`set\_config`

Merge key/value pairs into the server configuration

Parameters

- config (object, required) - Key/value pairs merged into the current configuration

`stream\_decode`

Run a sliding-window streaming decode session via backend.run_streaming_session

Parameters

- family (string, default 'rotated_surface') - One of: repetition, ring, rotated_surface, unrotated_surface, toric, heavy_hex, bicycle, bivariate_bicycle, hypergraph_product
- distance (integer, default 5) -
- window_size (integer, default 5) -
- n_rounds (integer, default 10) -
- error_rate (number, default 0.03) -
- seed (integer, default 1) -
- decoder_name (string, default 'union_find') - One of: union_find, fast_union_find, blossom, sparse_blossom, bp_osd, auto, hybrid, lookup_table, predecoded, auto_router, hybrid_cascade, gnn_belief_matching, belief_matching

`version\_info`

Live app + decoder-backend version report resolved from PyPI at call time (installed/local vs. latest, cached)

Parameters

- refresh (boolean, default False) - Bypass the version cache and re-query PyPI

MCP wire protocol

Newline-delimited JSON-RPC 2.0 over stdio. Launch with --mcp.

```
{ "jsonrpc": "2.0", "id": 1, "method": "initialize", "params": { "protocolVersion": "2024-11-05", "capabilities": {} }, "clientInfo": { "name": "mcp-client", "version": "0.1.0" } }
{ "jsonrpc": "2.0", "method": "notifications/initialized" }
{ "jsonrpc": "2.0", "id": 2, "method": "tools/list" }
{ "jsonrpc": "2.0", "id": 3, "method": "tools/call", "params": { "name": "decode_single", "arguments": { "family": "rotated_surface", "distance": 5, "window_size": 5, "n_rounds": 10, "error_rate": 0.03, "seed": 1, "decoder_name": "union_find" } } }
```

Result envelope: content[0].text holds a JSON payload; isError flags tool-level failure.

Common result schemas

```
# Single decode result
{"error":[...], "syndrome":[...], "result":{"correction":[...], "hamming_weight":int, "syndrome_valid":bool, "logical_failure":bool}
# Benchmark result
{"throughput_decodes_per_s":float, "decode_seconds":float, "n_trials":int, "p":float, "seed":int, "method":str, "backend":str}
# Resilient decode
{"success":bool, "used_decoder":str|None, "fallback_used":bool, "syndrome_valid":bool|None, "logical_failure":bool}
# Diagnostics report
{"overall_status":"pass|degraded|fail", "timestamp":float, "platform":str, "python":str, "workbench_version":str, "qvector_version":str}
```

Environment variables

```
| Variable | Effect |
|---|---|
| `QECTOR_PYTHON` | Compatible CPython for pip provisioning. |
| `QECTOR_DATA_DIR` | Relocate all QECTOR user data. |
| `QECTOR_AUTO_UPGRADE` | Set to `0` to disable background upgrade checks. |
| `QECTOR_APP_PACKAGE` | Override PyPI package name for app updates. |
| `QECTOR_SILENT` | Set to `1` to suppress the backend startup notice. |
| `QECTOR_LICENSE` | Ed25519 token that overrides academic/commercial for testing. |
```

Provisioning model

The Workbench application no longer bundles the decoder wheel. On first launch it downloads qector-decoder-v3 from PyPI and installs it into a managed, ABI-scoped user site. After installation the app works offline. Linux requires python3 and python3-pip; Windows and macOS need no separate Python setup.

Example workflows

```
import backend as be, autodebug
code = be.build_code("rotated_surface", 5)
out = be.run_single_decode(code, error_rate=0.05, decoder_kind="blossom", seed=42)
assert out["result"]["syndrome_valid"]

out2 = be.run_single_decode(code, error_rate=0.05, decoder_kind="bp_osd", seed=7, decoder_options={"bp_method": "union_find"})
probe = autodebug.probe_decoders("bivariate_bicycle", 3, seed=99)
resilient = autodebug.resilient_single_decode("bivariate_bicycle", 3, decoder="union_find", seed=7)
stats = be.run_hybrid_cascade_stats(code, n_samples=64, error_rate=0.05, seed=1)
```